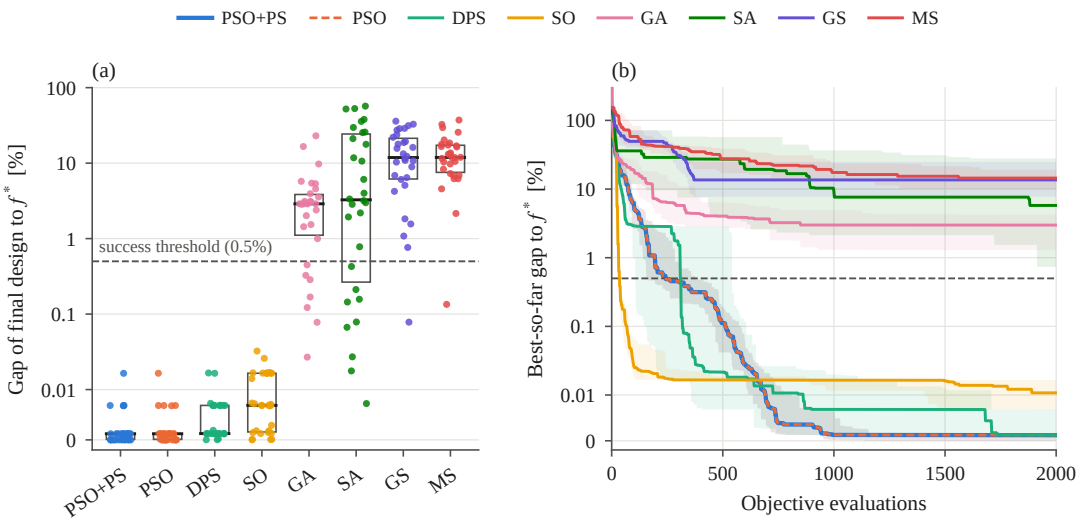

PSO+PS
PSO
DPS
SO
GA
SA
GS
MS
(a)
Gap of final design to f^* [%]
100
10
1
0.1
0.01
0
success threshold (0.5%)
PSO+PS
PSO
DPS
SO
GA
SA
GS
MS
(b)
Best-so-far gap to f^* [%]
100
10
1
0.1
0.01
0
0
500
1000
1500
2000
Objective evaluations